



PROBLEM SPACE

SOCIAL ROBOT CO-DESIGN CANVASES

What problem are you solving?

OU Staff spend significant time on routine vitals/reassessment; the robotic kiosk automates checks and flags deterioration early.

USER

Group(s)	Characteristics	Needs
Who are the users? Are there supporting users? For example: students and teachers. Observation Unit patients; Observation Unit nurses primary users secondary users Emergency physicians; clinical assistants; hospital administrators	What are the users like? Patients awaiting reassessment; nurses supervising multiple patients at once Responsible for clinical oversight; need reliable monitoring information	What do the users need? Easy interaction; safe measurements; clear instructions; fast deterioration alerts Timely alerts; accurate reassessment data; integration with hospital records
Goal(s)		
What do the primary and secondary users want to accomplish?		
Complete vitals; quick reassessment; flag urgent cases primary users secondary users Monitor condition efficiently; respond fast to alerts	Improve safety; reduce wait time; continuous monitoring Improve OU workflow; reduce staff workload; improve care quality	
SHORT-TERM		LONG-TERM

ROBOT

Task(s)			
What task does the robot perform?			
Guides patients through reassessment; collects vitals via sensors; runs the AI triage model; shows risk level.		Alerts nursing staff immediately on deterioration; builds a longitudinal monitoring history per patient.	
SHORT-TERM		LONG-TERM	
Advantages			
What advantages does using a robot bring (compared to a computer or human)?			
Speaks to patients; gives verbal instructions; offers reassurance during assessments	Encourages patient engagement; reduces anxiety while waiting	Adapts instructions to patient responses; supports multiple languages	Automated vital sign collection; standardised reassessment process
Social skills	User's emotional response	Personalization	Precise tasks
Heart rate; blood pressure; oxygen saturation; temperature	Positioned beside patient beds or within the Observation Unit	Guides patients to place arm/finger correctly for measurement	Connects to Mercer EHR; syncs monitors; alerts nursing stations; logs history
Data collection with sensors	Mobility	Environment manipulation	Connection to systems

